Airline Tweet Sentiment Analysis Using Text Mining and Machine Learning

Data Mining Project Report

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Course: Data Mining

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Note: Artificial intelligence was used in generating the code
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1. Dataset Description

This project uses the Tweets.csv dataset containing tweets directed at major U.S. airlines. Each observation represents one tweet and includes a sentiment label: negative, neutral, or positive. The dataset also contains useful contextual variables such as the airline name, sentiment confidence score, negative reason, tweet creation time, and retweet count.

The main variables used in the analysis are:

- airline_sentiment: target class to predict (negative, neutral, positive)
- airline_sentiment_confidence: confidence assigned to the sentiment label
- text: raw tweet content
- airline: airline mentioned in the tweet
- negativereason: specific reason when a tweet is negative
- tweet_created and retweet_count: contextual information used for exploration

The exploratory plots show a clear class imbalance: negative tweets dominate the dataset, which is an important issue for modeling and interpretation.

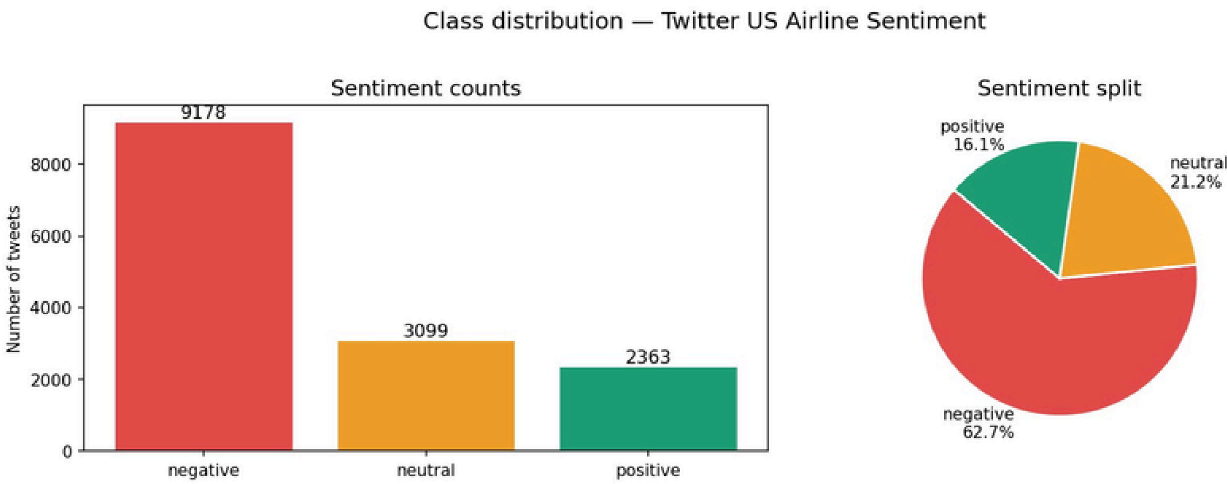

Figure 1. Overall class distribution of airline tweets.

2. Problem Statement

The goal of the project is to build a text-mining model that can classify airline tweets into negative, neutral, and positive sentiment categories. Beyond prediction, the project also aims to understand the main drivers of customer dissatisfaction and the language patterns associated with each sentiment class.

The main research questions are:

- Can we accurately classify tweet sentiment using text-mining methods?
- Which of the tested machine learning models performs best on this task?
- What words, airlines, and complaint categories are most associated with negative sentiment?

Target variable: airline_sentiment.

3. Data Preparation

The code first checks missing values and keeps only the variables relevant for analysis. Missing values are inspected as part of data-quality checking, while the selected variables focus the analysis on sentiment prediction and interpretation.

The following preprocessing steps were applied:

- Removal of low-confidence labels using a threshold of 0.60 on airline_sentiment_confidence.
- Selection of useful columns only: sentiment, confidence, text, airline, negative reason, tweet date, and retweet count.
- Text cleaning: lowercasing, removal of URLs, mentions, punctuation, HTML entities, and numbers.
- Expansion of contractions such as can't to cannot.
- Stopword removal while keeping important negation words such as not and never.
- Lemmatization to reduce word variation.
- Removal of tweets that became empty after cleaning.
- Label encoding of the target variable.
- Stratified split into train, validation, and test sets.

The confidence histogram below supports the filtering decision: most labels are concentrated near 1.0, and only a small fraction fall below the 0.60 threshold.

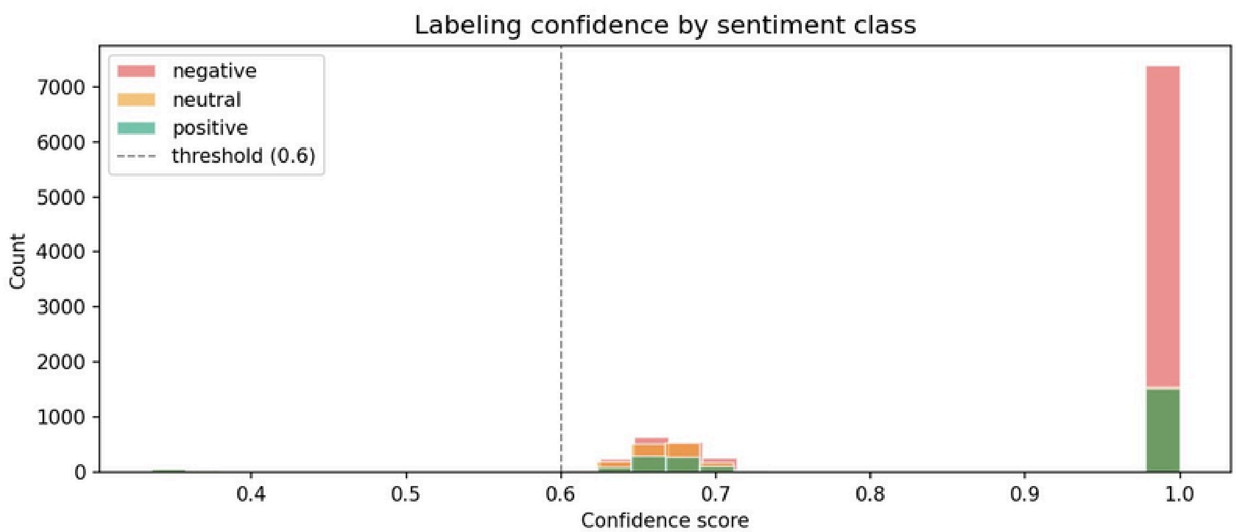

Figure 2. Distribution of label confidence by sentiment class.

4. Methods and Analysis

Because the problem is a text-classification task, the project uses text mining with TF-IDF features and compares three supervised learning models: Logistic Regression, Complement Naive Bayes, and Linear SVC.

Feature engineering and modeling pipeline:

- TF-IDF vectorization with unigrams and bigrams.

- Limiting the vocabulary to the most informative features and removing overly rare or overly common terms.
- Creation of hand-crafted features from the raw tweet, including character count, word count, punctuation counts, emoji presence, hashtag count, mention count, and a negation indicator.
- Standardization of hand-crafted features before combining them with TF-IDF features.
- Use of class_weight="balanced" for Logistic Regression and Linear SVC to reduce the effect of class imbalance.
- 5-fold cross-validation on the training data and comparison on the validation set before choosing the final model.

Why these models? Naive Bayes is a classical baseline for text classification, Logistic Regression is a strong linear model for sparse text features, and Linear SVC is widely known to perform well on high-dimensional TF-IDF data.

5. Results and Interpretation

5.1 Exploratory findings

The dataset is heavily dominated by negative sentiment. As shown in Figure 1, 62.7% of tweets are negative, compared with 21.2% neutral and 16.1% positive. This confirms that class imbalance must be considered when building the models.

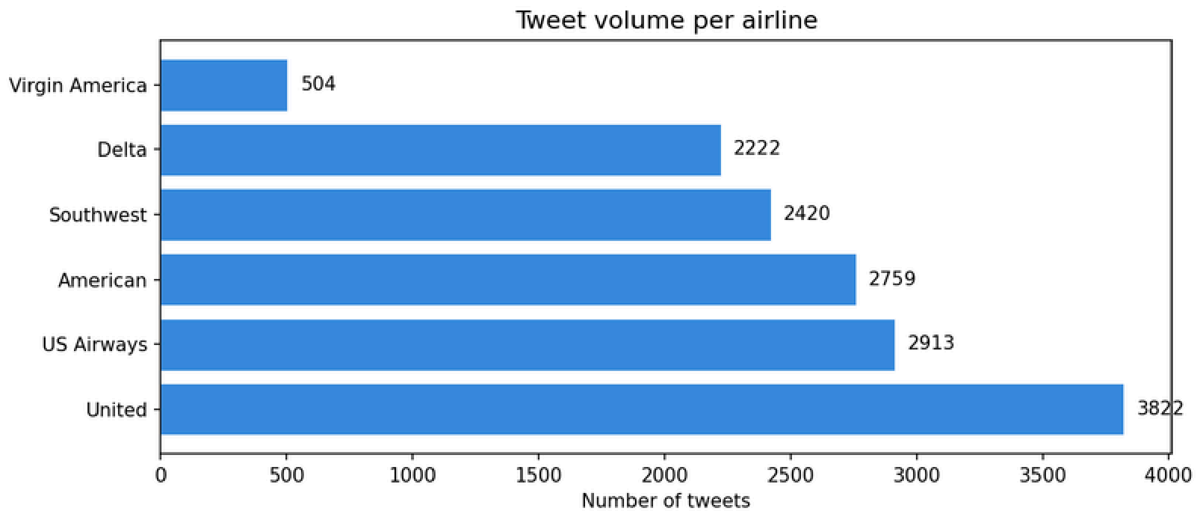

Figure 3. Number of tweets by airline.

United, US Airways, and American generate the largest tweet volumes, while Virgin America has the smallest volume. This matters because airlines with more tweets may contribute more heavily to the vocabulary seen by the models.

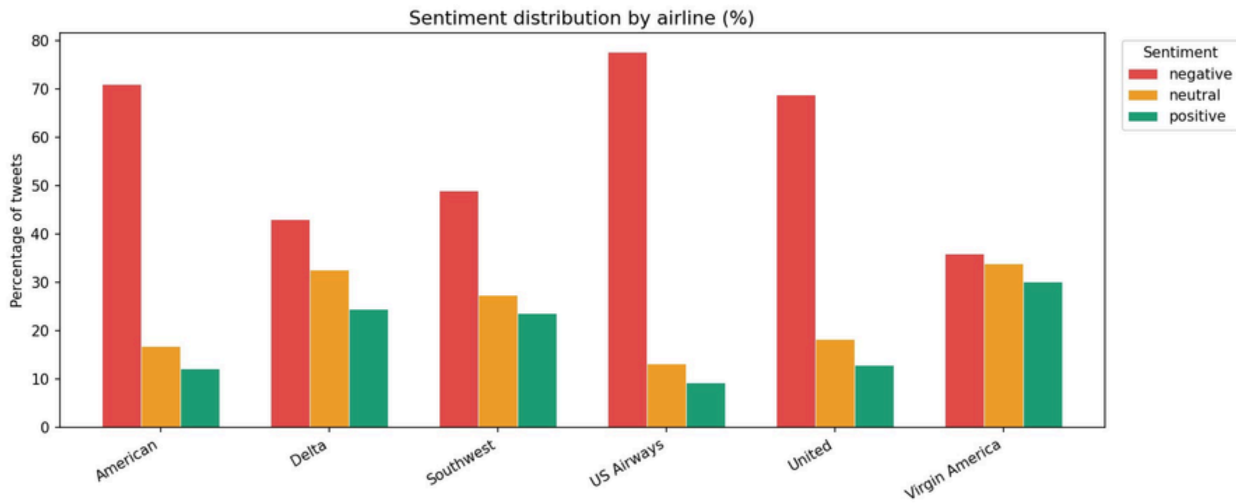

Figure 4. Sentiment distribution by airline.

US Airways has the highest percentage of negative tweets, followed by American and United. Virgin America is the most balanced airline and has the largest positive share. This suggests important differences in customer experience and public perception across airlines.

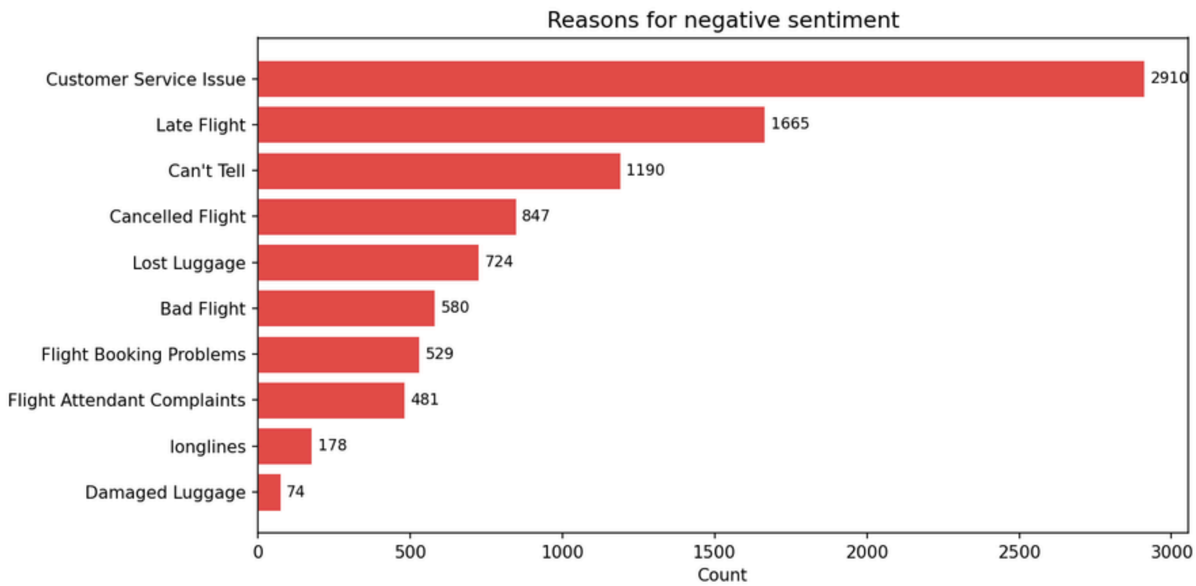

Figure 5. Main reasons behind negative sentiment.

Customer Service Issue is by far the most frequent complaint, followed by Late Flight, Can't Tell, and Cancelled Flight. This indicates that customer service and operational disruptions are the main drivers of dissatisfaction in the dataset.

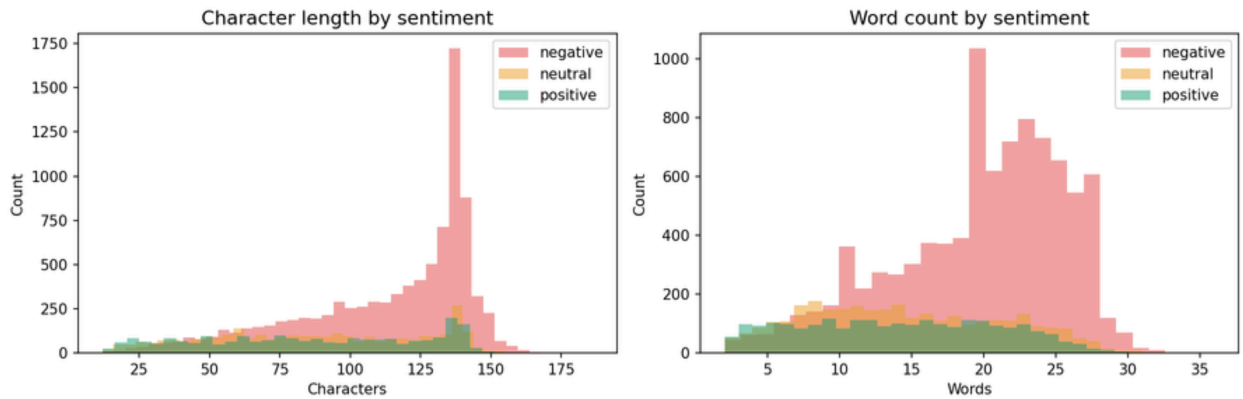

Figure 6. Character length and word count by sentiment class.

Negative tweets tend to be longer and contain more words than neutral and positive tweets. This pattern suggests that dissatisfied customers often provide more detailed explanations or complaints.

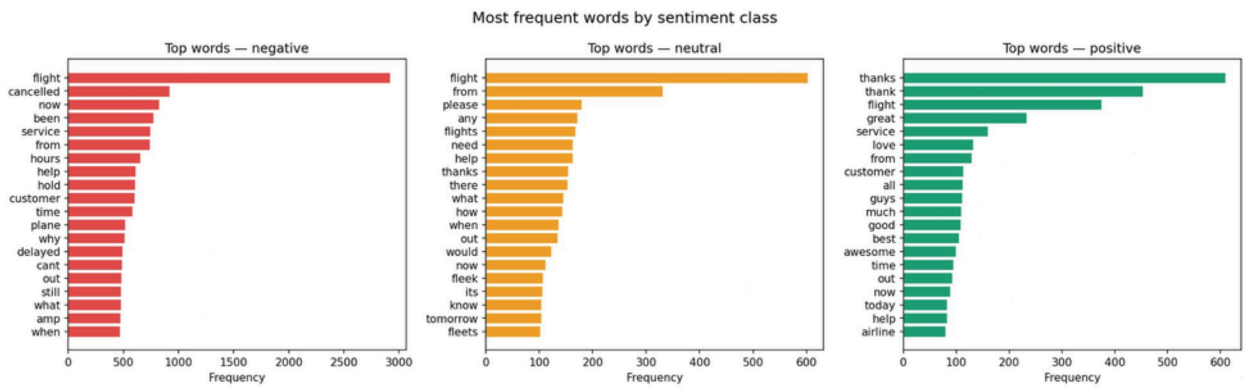

Figure 7. Most frequent words by sentiment class.

The word patterns are consistent with the sentiment labels. Negative tweets are dominated by words such as cancelled, service, hours, help, and delayed. Neutral tweets mainly contain request-like or informational words such as please, need, and when. Positive tweets are centered around gratitude and praise, especially thanks, thank, great, love, and awesome.

5.2 Model comparison on the validation set

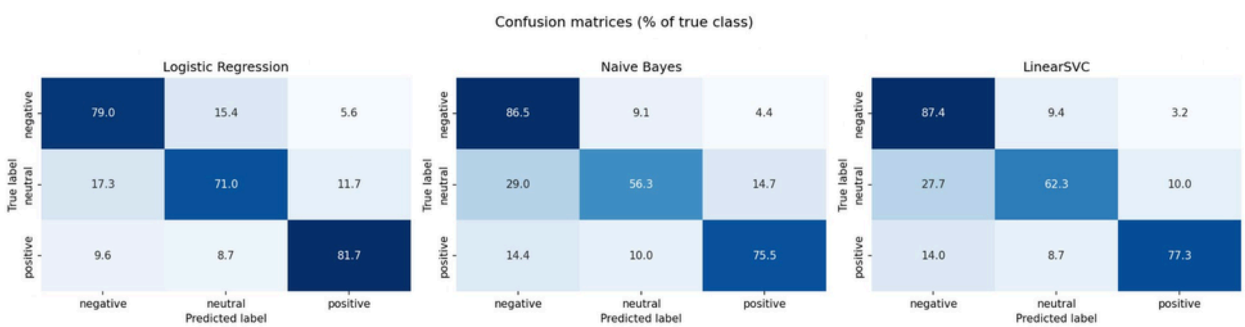

Figure 8. Validation confusion matrices for the three models.

All three models identify negative tweets relatively well, but neutral tweets remain the most difficult class. Logistic Regression gives the highest neutral recall, while Linear SVC provides the most balanced behavior across the three classes. Naive Bayes remains strong on negative tweets but weaker on neutral ones.

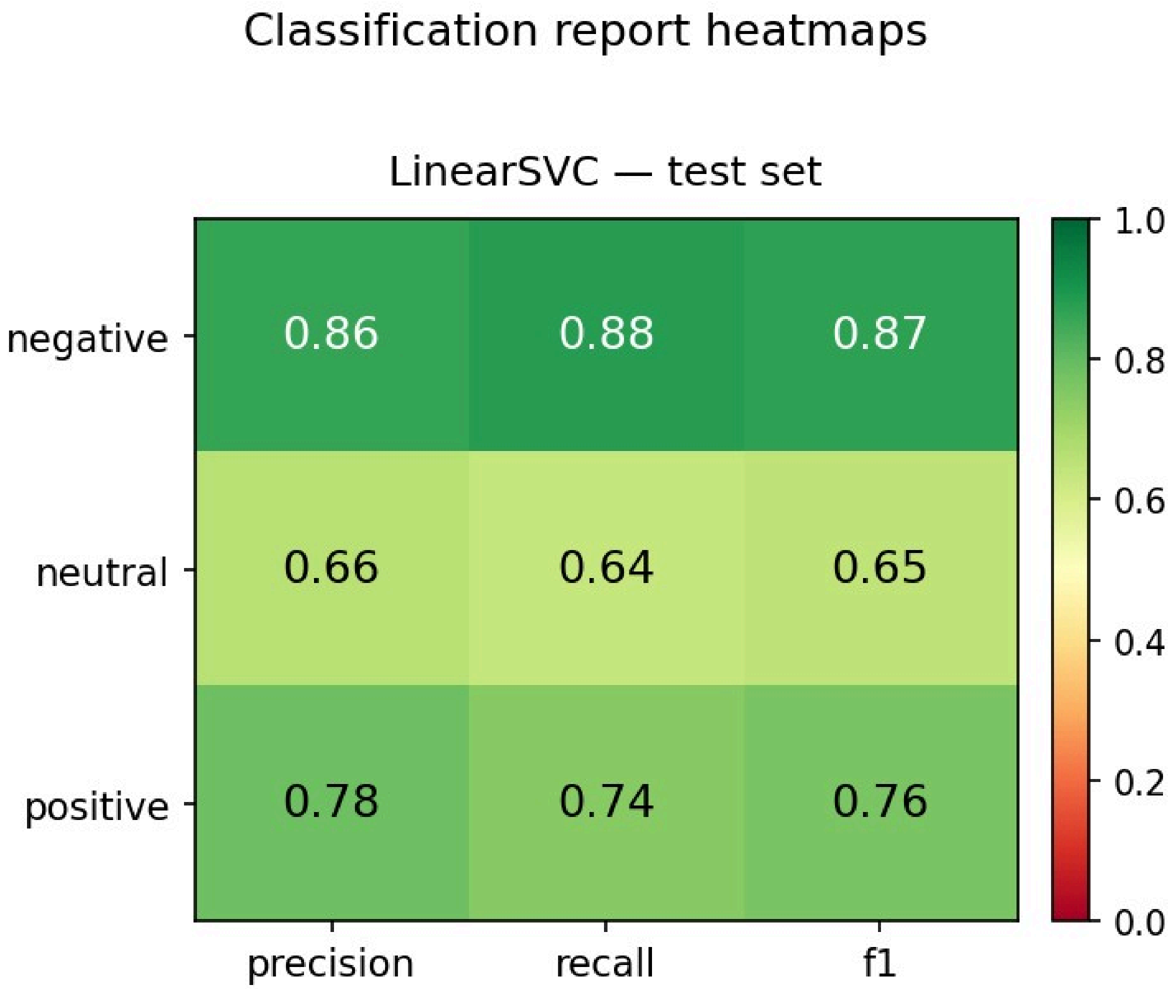

Figure 9. Final classification report heatmap for Linear SVC on the test set.

The final classification report for Linear SVC confirms that negative sentiment is the easiest class and neutral is the hardest. The model achieves strong performance on negative tweets, reasonable performance on positive tweets, and lower performance on neutral tweets, which remains the most challenging class.

5.3 Final test-set performance

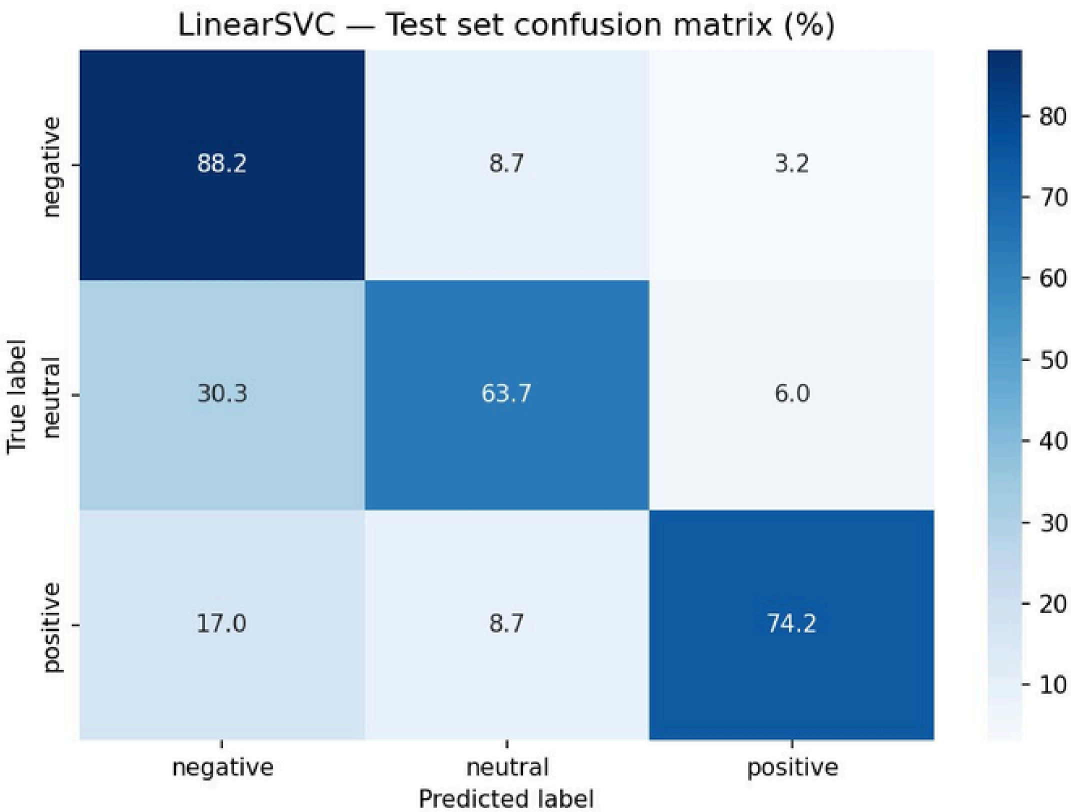

Figure 10. Linear SVC confusion matrix on the test set.

On the test set, Linear SVC correctly classifies 88.2% of negative tweets, 63.7% of neutral tweets, and 74.2% of positive tweets. The main weakness remains the confusion between neutral and negative tweets: 30.3% of neutral tweets are predicted as negative. This shows that neutral airline tweets often contain complaint-like vocabulary even when they are not truly negative.
The final classification report for Linear SVC on the test set is summarized below.

Class	Precision	Recall	F1-score
negative	0.86	0.88	0.87
neutral	0.66	0.64	0.65
positive	0.78	0.74	0.76

6. Conclusion

In this project, airline tweets were analyzed using text-mining techniques to classify sentiment into negative, neutral, and positive categories. After cleaning the text and transforming it into TF-IDF and hand-crafted features, three machine learning models were compared: Logistic Regression, Complement Naive Bayes, and Linear SVC.

The results showed that negative tweets are the easiest to identify, while neutral tweets are the most difficult because they are often confused with negative ones. Overall, Linear SVC provided the best balance across the three classes and was therefore selected as the final model. The exploratory analysis also revealed that customer service issues, late flights, and cancellations are the main sources of negative sentiment, whereas positive tweets are mainly associated with gratitude and satisfaction. This project demonstrates that text mining can effectively extract useful customer insights and support sentiment prediction in airline-related social media data.